Assignment (Term Paper)

Statistical Programming, WS Term 2025/2026

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Section 1: Data Wrangling and Spatial Exploration

This notebook covers the cleaning, transformation, and first exploration of the Berlin Airbnb data. The aim is to prepare a usable dataset and describe the main price patterns in a clear way.

```
# Task 1.1

import numpy as np
import pandas as pd

listings = pd.read_csv("../Daten/listings_berlin.csv")

print(f"Number of rows: {len(listings)}")
print(f"Number of columns: {len(listings.columns)}")
missing_count = listings.isna().sum()
missing_share = listings.isna().mean().map("{:.2%}".format)

Table_1_1 = pd.DataFrame({"Number of missings": missing_count, "Share of missings": missing_share})
Table_1_1.head(20)
```

```
Number of rows: 14274
Number of columns: 20
```

	Number of missings	Share of missings
price	5010	35.10%
neighbourhood_cleansed	0	0.00%
latitude	0	0.00%
longitude	0	0.00%
room_type	0	0.00%
bedrooms	2024	14.18%
bathrooms_text	16	0.11%
accommodates	0	0.00%
amenities	0	0.00%
host_since	14	0.10%
host_is_superhost	164	1.15%
host_listings_count	14	0.10%
host_response_rate	5018	35.15%
review_scores_rating	3314	23.22%
review_scores_cleanliness	3314	23.22%
number_of_reviews	0	0.00%
reviews_per_month	3314	23.22%
minimum_nights	0	0.00%
availability_365	0	0.00%
instant_bookable	0	0.00%

```
# Task 1.2
# Sort the table by missing-value share and display the top three columns.
print("The following three variables have the highest missing-value rates:")
Table_1_1.sort_values("Share of missings", ascending=False).head(3)
```

```
The following three variables have the highest missing-value rates:
```

	Number of missings	Share of missings
host_response_rate	5018	35.15%
price	5010	35.10%
reviews_per_month	3314	23.22%

The three variables with the largest missing-value shares need different treatment.

- **Price** should clearly be kept, since it is the main variable for later analysis. Missing prices cannot be imputed in a convincing way here, so we keep the column and only drop observations when price is required in a specific step.
- **Host response rate** may still contain useful host-level information. We therefore keep the column for now instead of dropping it immediately.
- **Reviews per month** is less essential because the dataset already includes the total number of reviews. For our later analysis it is not central, so this variable could be dropped without losing much information.

Overall, we prefer to keep potentially relevant variables unless the later analysis shows that they are not needed.

```
# Task 1.3
# Convert the price column from string format to numeric values.
def clean_price(price_str):
    """Convert a price string to a numeric value."""

    if pd.isna(price_str):
        return np.nan
    price_str_clean = str(price_str).replace("$", "").replace(',', '')
    try:
        price_number = float(price_str_clean)
        if price_number > 0:
            return price_number
        return np.nan
    except ValueError:
        return np.nan

# Apply the cleaning function to the full price column.
listings_1_3 = listings.copy()
listings_1_3["price"] = listings_1_3["price"].apply(clean_price)
listings_1_3["price"].describe().loc[['min', '50%', 'mean', 'max']]
```

```
min         5.000000
50%       104.000000
mean      201.240393
max     50000.000000
Name: price, dtype: float64
```

```
# Task 1.4
listings_1_4 = listings_1_3.copy()
# We first inspect the unique entries in bathrooms_text if needed.
# listings_1_4["bathrooms_text"].unique()

# Convert the bathroom information from text into a numeric representation.
# Rule 1: "private" is redundant for the numeric count.
# Rule 2: "shared" is stored in a separate boolean column.
# Rule 3: a half bath is coded as 0.5.

# Store shared-bath information.
listings_1_4["Shared_bath_status"] = listings_1_4["bathrooms_text"].str.contains("shared", case=Falsc
# Recode half baths.
listings_1_4["bathrooms_text"] = listings_1_4["bathrooms_text"].str.replace("Half-Bath", "0.5", case=
# Convert the remaining text to numeric values.
listings_1_4["bathrooms_text"] = pd.to_numeric(
    listings_1_4["bathrooms_text"].str.replace({
        "baths": "",
        "shared": "",
        "private": "",
        "bath": "",
```

```
        " ": ""
    }, case=False)
)
listings_1_4.rename(columns={"bathrooms_text": "bathrooms_number"}, inplace=True)
listings_1_4[["bathrooms_number", "Shared_bath_status"]].head()
print(listings_1_4.isna().sum())
```

```
price                        5010
neighbourhood_cleansed          0
latitude                        0
longitude                       0
room_type                       0
bedrooms                     2024
bathrooms_number               16
accommodates                    0
amenities                       0
host_since                     14
host_is_superhost             164
host_listings_count            14
host_response_rate           5018
review_scores_rating         3314
review_scores_cleanliness    3314
number_of_reviews               0
reviews_per_month            3314
minimum_nights                  0
availability_365                0
instant_bookable                0
Shared_bath_status              0
dtype: int64
```

```
# Task 1.5
listings_1_5 = listings_1_4.copy()
# Convert "t" and "f" to boolean values.
listings_1_5["instant_bookable"] = listings_1_5["instant_bookable"] == "t"
listings_1_5["host_is_superhost"] = listings_1_5["host_is_superhost"] == "t"
# Create host tenure in years and drop the original date column.
listings_1_5["host_tenure_years"] = 2025 - pd.to_datetime(listings_1_5["host_since"]).dt.year
listings_1_5 = listings_1_5.drop(columns="host_since")
listings_1_5.columns
```

```
Index(['price', 'neighbourhood_cleansed', 'latitude', 'longitude', 'room_type',
       'bedrooms', 'bathrooms_number', 'accommodates', 'amenities',
       'host_is_superhost', 'host_listings_count', 'host_response_rate',
       'review_scores_rating', 'review_scores_cleanliness',
       'number_of_reviews', 'reviews_per_month', 'minimum_nights',
       'availability_365', 'instant_bookable', 'Shared_bath_status',
       'host_tenure_years'],
      dtype='str')
```

```
# Task 1.6
import matplotlib.pyplot as plt
import seaborn as sns

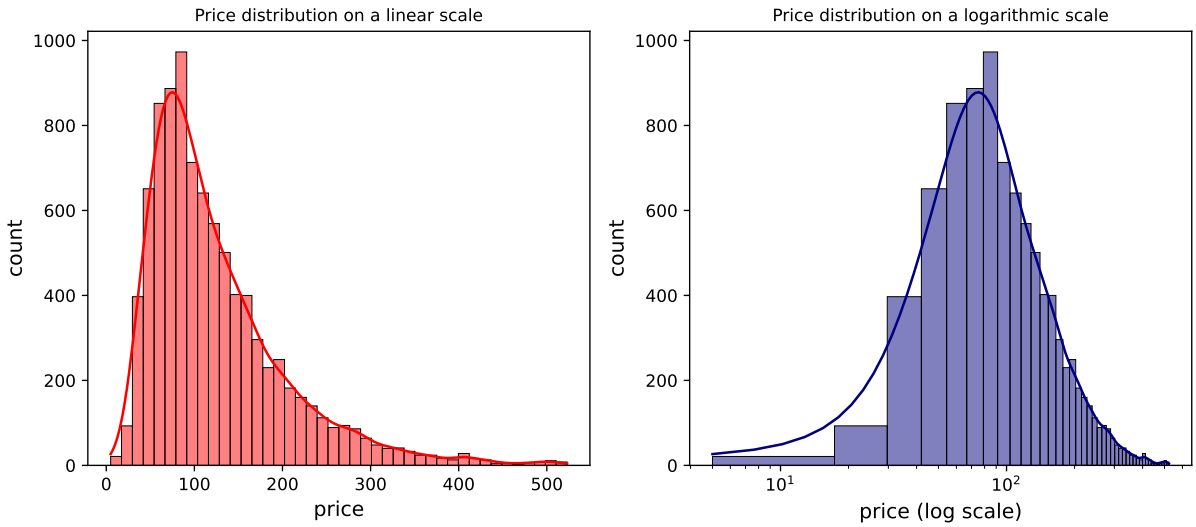
log_prices = np.log(listings_1_5["price"])
upper_bound = np.exp(np.mean(log_prices) + 2.33 * np.std(log_prices))
# A z-value of 2.33 covers about 99% of the log-price distribution.

listings_1_6 = listings_1_5.loc[listings_1_5["price"] <= upper_bound].copy()
```

```
fig, ax = plt.subplots(1, 2, sharey=False, figsize=(10, 4.5))

sns.histplot(data=listings_1_6, x="price", kde=True, ax=ax[0], log=False,
             bins=round(10*np.log10(len(listings_1_5))), color="red", stat="count")
ax[0].set_xlabel("price", fontsize=12)
ax[0].set_ylabel("count", fontsize=12)
ax[0].set_title("Price distribution on a linear scale", fontsize=10)

sns.histplot(data=listings_1_6, x="price", kde=True, ax=ax[1], log=False,
             bins=round(10*np.log10(len(listings_1_5))), color="navy", stat="count")
ax[1].set_xlabel("price (log scale)", fontsize=12)
ax[1].set_ylabel("count", fontsize=12)
ax[1].set_xscale("log")
ax[1].set_title("Price distribution on a logarithmic scale", fontsize=10)
plt.tight_layout()
plt.show()
```



The histogram on the linear scale is heavily dominated by a small number of very expensive listings, so most of the distribution is compressed into a narrow range. The logarithmic version is more informative because it spreads out the lower and middle price ranges and makes the overall shape easier to read.

For later statistical analysis, the logarithmic representation is the better choice. Prices appear roughly log-normal, and the log scale reduces the influence of extreme values. This makes the distribution more stable and better suited for regression-based work.

```
# Task 1.7
# Group by neighbourhood and compute summary statistics for price and ratings.
table_1_7 = listings_1_5.groupby("neighbourhood_cleansed").agg({
    "price": ["mean", "median", "count"],
    "review_scores_rating": "mean"
}).copy()

# Round selected values and sort by median price.
table_1_7 = table_1_7.round({("price", "mean"): 1, ("review_scores_rating", "mean"): 1})

table_1_7.head(3)
# The three most expensive areas in this table are Haselhorst, West 5, and Schmargendorf.
```

neighbourhood_cleansed	price mean	median	count	review_scores_rating mean
Haselhorst	216.2	236.5	6	4.9
West 5	381.5	203.0	11	4.8
Schmargendorf	170.6	172.0	34	4.7

```
# Task 1.8
listings_1_8 = listings_1_5.copy()

def trimmed_mean(x, pct):
    """Calculate the trimmed mean after removing pct percent from both tails."""
    if not (isinstance(pct, (int, float)) and 0 <= pct < 50):
        raise ValueError("pct must be a number between 0 and 50")
    try:
        x = np.asarray(x, dtype=float)
    except (TypeError, ValueError):
        raise TypeError("x must be an array-like object containing numeric values")
    if x.ndim != 1:
        raise ValueError("x must be one-dimensional")
    bound = int(len(x) * pct / 100)
    if bound == 0:
        return np.mean(x)
    if 2 * bound >= len(x):
        raise ValueError("pct too large for the size of x")
    x = x[~np.isnan(x)]
    x = np.sort(x)
    mean_trimmed = x[bound:-bound]
    return np.mean(mean_trimmed)

price_mean_trimmed = trimmed_mean(listings_1_8["price"], 5)
price_mean = listings_1_8["price"].mean()
print(f"The 5% trimmed mean of price is: {price_mean_trimmed:.2f}. The standard mean of price is: {pr

print("The trimmed mean is lower because a small number of very expensive listings pull the standard
```

```
The 5% trimmed mean of price is: 116.85. The standard mean of price is: 201.24. This is an increase o
The trimmed mean is lower because a small number of very expensive listings pull the standard mean up
```

```
import ast

# Parse the amenities column from string representation to a Python list.
def clean_amenities(amenities_str):
    if pd.isna(amenities_str):
        return []
    try:
        amenities = ast.literal_eval(amenities_str)
        return [a.strip() for a in amenities if isinstance(a, str)]
    except (ValueError, SyntaxError):
        return []

listings_1_8["amenities_list"] = listings_1_8["amenities"].apply(clean_amenities)
listings_1_8["amenities_count"] = listings_1_8["amenities_list"].apply(len)
listings_1_8[["amenities", "amenities_list", "amenities_count"]].head()

print(listings_1_8["amenities_count"].sum())
```

```
379295
```

```
# Task 1.9
listings_1_5["amenities_list"] = listings_1_5["amenities"].apply(clean_amenities)
listings_1_5["amenities_count"] = listings_1_5["amenities_list"].apply(len)

listings_1_9 = listings_1_5.copy()

fig, ax = plt.subplots(2, 2, figsize=(11, 7), dpi=200)

sns.violinplot(data=listings_1_9, x="room_type", y="price", hue="Shared_bath_status", log_scale=True
ax[0,0].set_title("Price distribution by room type and shared bath status", fontsize=10)
ax[0,0].set_xlabel("room type", fontsize=10)
ax[0,0].set_ylabel("price (log scale)", fontsize=10)

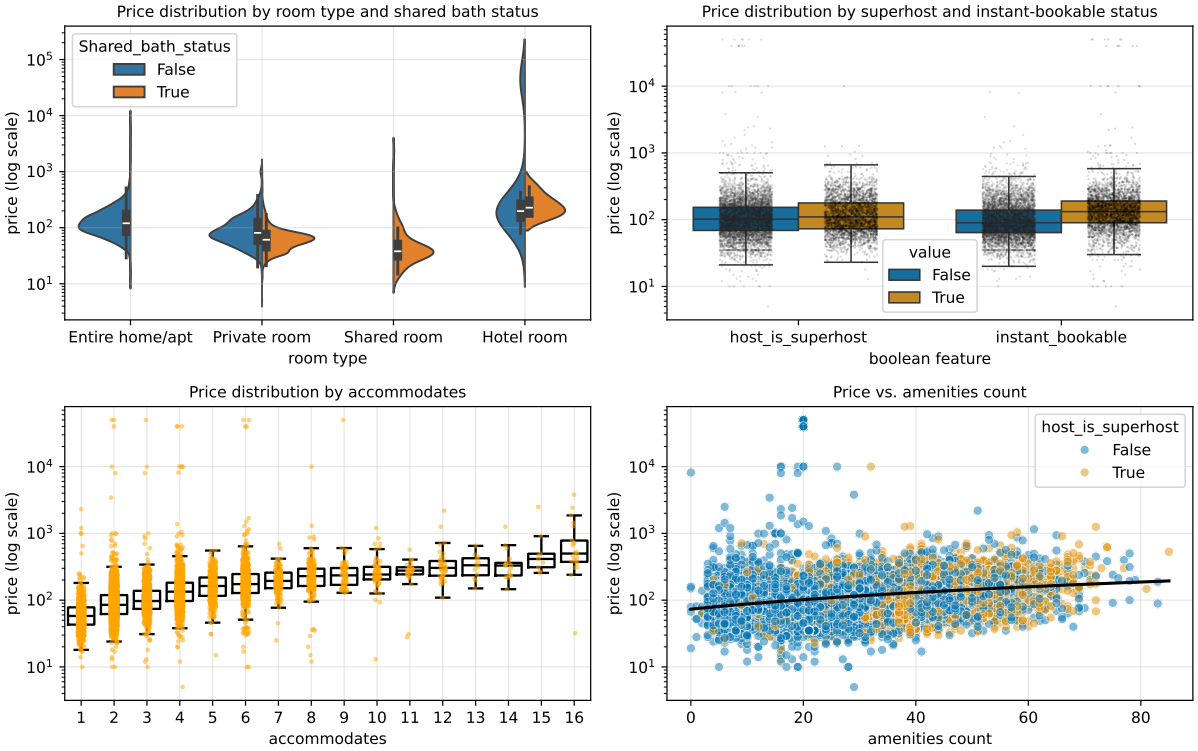
df_long = listings_1_9.melt(
    id_vars=["price"],
    value_vars=["host_is_superhost", "instant_bookable"],
    var_name="feature",
    value_name="value")
sns.boxplot(data=df_long, x="feature", y="price", hue="value", showfliers=False, palette="colorblind
sns.stripplot(data=df_long, x="feature", y="price", hue="value", dodge=True, alpha=0.15, size=1.5, ji
              palette="dark:black", ax=ax[0,1], log_scale=True, legend=False)
ax[0,1].set_title("Price distribution by superhost and instant-bookable status", fontsize=10)
ax[0,1].set_xlabel("boolean feature", fontsize=10)
ax[0,1].set_ylabel("price (log scale)", fontsize=10)

sns.boxplot(data=listings_1_9, x="accommodates", y="price", ax=ax[1,0], log_scale=True, color="black
sns.stripplot(data=listings_1_9, x="accommodates", y="price", ax=ax[1,0], alpha=0.5, log_scale=True,
ax[1,0].set_title("Price distribution by accommodates", fontsize=10)
ax[1,0].set_xlabel("accommodates", fontsize=10)
ax[1,0].set_ylabel("price (log scale)", fontsize=10)

sns.scatterplot(data=listings_1_9, x="amenities_count", y="price", ax=ax[1,1], alpha=0.5, hue="host_
sns.regplot(data=listings_1_9, x="amenities_count", y="price", ax=ax[1,1], scatter=False, line_kws={
ax[1,1].set_title("Price vs. amenities count", fontsize=10)
ax[1,1].set_yscale("log")
ax[1,1].set_xlabel("amenities count", fontsize=10)
ax[1,1].set_ylabel("price (log scale)", fontsize=10)

for a in ax.flatten():
    a.grid(alpha=0.3)
ax[0,1].xaxis.grid(False)

plt.tight_layout()
plt.show()
```




```
print("Hotel room shared bath rate:",(listings_1_9.loc[listings_1_9["room_type"]=="Hotel room"]["Sha
print("Private room shared bath rate:",(listings_1_9.loc[listings_1_9["room_type"]=="Private room"][
```

```
Hotel room shared bath rate: 0.05454545454545454
Private room shared bath rate: 0.5687656285519436
```

The first plot shows large price differences across room types. Shared rooms are clearly the cheapest, private rooms come next, and entire homes or hotel rooms tend to be more expensive. Shared bathrooms are mainly relevant for private rooms and hotel rooms, and listings with shared bathrooms are usually priced lower.

The second plot suggests that both superhost status and instant booking are associated with higher prices. The pattern is visible in both boxplots, although the differences are not equally strong.

The third plot shows a clear positive relationship between the number of accommodated guests and price. Larger listings are generally more expensive.

The fourth plot indicates a positive relationship between the number of amenities and price. The trend is not perfect, but the fitted line points upward, which suggests that more heavily equipped listings tend to charge more.

```
# Run with the "" counting method used for 1_9 and the ast method for 1_8
n_1_8 = 0
n_1_9 = 0
for idx in listings_1_8.index:
    n_1_8 += listings_1_8.loc[idx, "amenities_count"]
    n_1_9 += listings_1_9.loc[idx, "amenities_count"]
print(f"Total amenities count in listings_1_8: {n_1_8}")
print(f"Total amenities count in listings_1_9: {n_1_9}")
print(listings_1_8["amenities_count"].sum())
print(listings_1_9["amenities_count"].sum())
print(listings_1_9.loc[listings_1_9["price"] > upper_bound, "amenities_count"].sum())
```

```
Total amenities count in listings_1_8: 379295
Total amenities count in listings_1_9: 379295
379295
379295
```

3695

```
# Optional extra plots for Section 1
# These plots are less central for the final discussion but still useful for exploration.

fig, ax = plt.subplots(2, 2, figsize=(11, 7), dpi=200)

sns.scatterplot(data=listings_1_9, x="review_scores_rating", y="number_of_reviews", ax=ax[0,0], alpha
ax[0,0].set_title("Review scores vs. number of reviews", fontsize=10)
ax[0,0].set_yscale("log")
ax[0,0].set_xlabel("review scores", fontsize=10)
ax[0,0].set_ylabel("number of reviews (log scale)", fontsize=10)

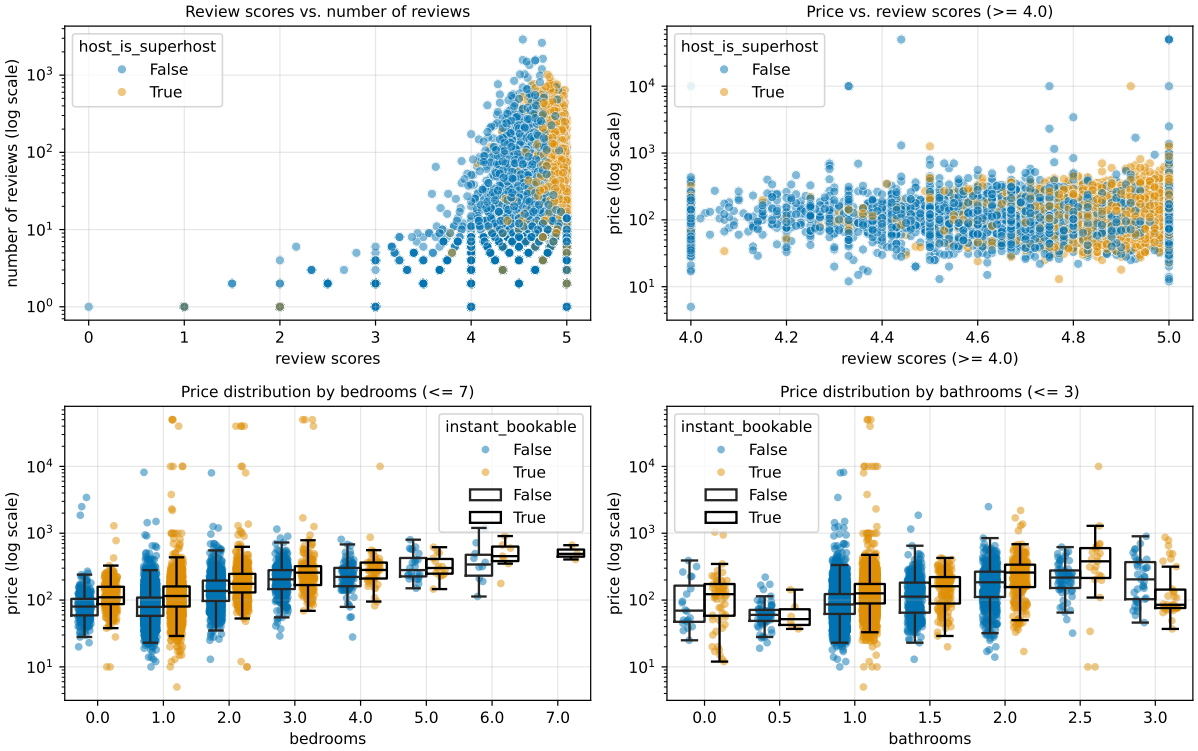
sns.scatterplot(data=listings_1_9, x=listings_1_9.loc[(listings_1_9["review_scores_rating"] >= 4.0)]
ax[0,1].set_yscale("log")
ax[0,1].set_title("Price vs. review scores (>= 4.0)", fontsize=10)
ax[0,1].set_xlabel("review scores (>= 4.0)", fontsize=10)
ax[0,1].set_ylabel("price (log scale)", fontsize=10)

sns.stripplot(data=listings_1_9.loc[(listings_1_9["bedrooms"] <= 7)], x="bedrooms", y="price", ax=ax
sns.boxplot(data=listings_1_9.loc[(listings_1_9["bedrooms"] <= 7)], x="bedrooms", y="price", ax=ax[1
ax[1,0].set_title("Price distribution by bedrooms (<= 7)", fontsize=10)
ax[1,0].set_xlabel("bedrooms", fontsize=10)
ax[1,0].set_ylabel("price (log scale)", fontsize=10)

sns.stripplot(data=listings_1_9.loc[(listings_1_9["bathrooms_number"] <= 3)], x="bathrooms_number", 
sns.boxplot(data=listings_1_9.loc[(listings_1_9["bathrooms_number"] <= 3)], x="bathrooms_number", y=
ax[1,1].set_title("Price distribution by bathrooms (<= 3)", fontsize=10)
ax[1,1].set_xlabel("bathrooms", fontsize=10)
ax[1,1].set_ylabel("price (log scale)", fontsize=10)

for a in ax.flatten():
    a.grid(alpha=0.3)

plt.tight_layout()
plt.show()
```



The Berlin Airbnb market shows a clear price gradient across listing characteristics. Room type is one of the strongest factors: shared rooms are cheapest, while entire homes and hotel rooms are more expensive. Capacity also matters a lot, since listings for more guests tend to have substantially higher prices. In addition, listings with more amenities are usually more expensive, which suggests that quality and equipment are reflected in pricing. Superhost status and instant booking also seem to be associated with a price premium, although these effects are less pronounced than room type or capacity. Spatially, central areas often combine high listing density with comparatively high prices, but the relationship is not perfect. Some peripheral regions also show high median prices, likely because they contain fewer listings and are therefore more sensitive to individual outliers. Overall, listing size, type, amenities, and location seem to matter most for prices.

```
listings_1_9[["bedrooms", "room_type","amenities_count"]].head()
```

	bedrooms	room_type	amenities_count
0	1.0	Entire home/apt	44
1	4.0	Entire home/apt	32
2	0.0	Entire home/apt	14
3	0.0	Entire home/apt	18
4	2.0	Entire home/apt	30

```
# Task 1.11

import json

with open("../Daten/berlin_bezirke.geojson", "r", encoding="utf-8") as f:
    berlin_geojson = json.load(f)

print(len(berlin_geojson["features"]))
berlin_geojson["features"][0].keys()
```

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143
```

```
dict_keys(['type', 'properties', 'geometry', 'id'])
```

```python
# Task 1.11: match region information from the Airbnb data to the GeoJSON data

import json
import re

with open("../Daten/berlin_bezirke.geojson", "r", encoding="utf-8") as f:
    berlin_geojson = json.load(f)

print(len(berlin_geojson["features"]))
berlin_geojson["features"][0].keys()

bzr_names = [feature["properties"]["BZR_NAME"] for feature in berlin_geojson["features"]]
geo_df = pd.DataFrame({"region_geo": bzr_names})

airbnb_regions = listings_1_9["neighbourhood_cleansed"].unique()
airbnb_df = pd.DataFrame({"region_airbnb": airbnb_regions})

def clean_region(name):
    name = name.lower()
    name = name.replace("str.", "strasse")
    name = name.replace("-", " ")
    name = name.replace("ß", "ss")
    name = name.replace(" / ", "/")
    name = re.sub(r"\s+", " ", name)
    name = name.strip()

    name = name.split("/")[0]
    name = re.sub(r"(west \d+).*", lambda m: m.group(1), name)
    name = re.sub(r"(nord \d+).*", lambda m: m.group(1), name)
    name = re.sub(r"(ost \d+).*", lambda m: m.group(1), name)

    return name

geo_df["clean_name"] = geo_df["region_geo"].apply(clean_region)
airbnb_df["clean_name"] = airbnb_df["region_airbnb"].apply(clean_region)

auto_matches = set(geo_df["clean_name"]) & set(airbnb_df["clean_name"])
mapping_auto = {name: name for name in auto_matches}
mapping_auto_v2 = {
    airbnb_df.loc[airbnb_df["clean_name"] == name]["region_airbnb"].iloc[0]:
    geo_df.loc[geo_df["clean_name"] == name]["region_geo"].iloc[0]
    for name in auto_matches
}

remaining_airbnb = set(airbnb_df["clean_name"]) - set(mapping_auto.keys())
remaining_geo = set(geo_df["clean_name"]) - set(mapping_auto.values())
print("Remaining Airbnb regions:", sorted(remaining_airbnb))
print("Remaining GeoJSON regions:", sorted(remaining_geo))

# For the unmatched names below, we use a manual mapping after checking the cleaned names.
mapping_manual = {
    "friedenau": "friedenau ost",
    "gropiusstadt": "gropiusstadt ost",
    "mariendorf": "mariendorf nord",
    "marienfelde": "marienfelde nord",
    "lichtenrade": "lichtenrade nord",
    "tempelhof": "tempelhof nord",
    "mv 1": "mv nord",
```

```
    "mv 2": "mv süd",
    "schöneberg nord": "schöneberg nordost",
    "schöneberg süd": "schöneberg südost",
    "kölln. vorstadt": "köllnische vorstadt",
}

mapping_manual_v2 = {
    airbnb_df.loc[airbnb_df["clean_name"] == name_a]["region_airbnb"].iloc[0]:
    geo_df.loc[geo_df["clean_name"] == name_g]["region_geo"].iloc[0]
    for name_a, name_g in mapping_manual.items()
}

mapping = {**mapping_auto_v2, **mapping_manual_v2}
print(mapping)
```

```
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Remaining Airbnb regions: ['altstadt kietz', 'barstrasse', 'düsseldorfer strasse', 'forst grunewald'
Remaining GeoJSON regions: ['altstadt köpenick', 'britz nord', 'friedenau ost', 'friedenau west', 'gr
{'Adlershof': 'Adlershof', 'Weißensee Ost': 'Weißensee Ost', 'Gatow / Kladow': 'Gatow/Kladow', 'Marza
Mitte': 'Marzahn Mitte', 'Niederschöneweide': 'Niederschöneweide', 'Pankow Zentrum': 'Pankow Zentrum
Marx-Allee-Nord': 'Karl-Marx-Allee Nord', 'Frankfurter Allee Nord': 'Frankfurter Allee Nord', 'Otto-
Suhr-Allee': 'Otto-Suhr-Allee/Kantstraße', 'Schönholz/Wilhelmsruh/Rosenthal': 'Schönholz/Wilhelmsruh/
Hohenschönhausen Süd': 'Alt-Hohenschönhausen Süd', 'Schillerpromenade': 'Schillerpromenade', 'Heller
Süd': 'Hellersdorf Süd', 'Weißensee': 'Weißensee', 'Köpenick-Süd': 'Köpenick Süd', 'Müggelheim': 'Mü
Süd/Flughafensee', 'Drakestr.': 'Drakestraße', 'Grunewald': 'Grunewald', 'Blankenfelde/Niederschönhau
Ost': 'Hellersdorf Ost', 'Buckow Nord': 'Buckow Nord', 'Rummelsburger Bucht': 'Rummelsburger Bucht',
Hohenschönhausen Nord': 'Alt-Hohenschönhausen Nord', 'nördliche Luisenstadt': 'Nördliche Luisenstadt
Lichtenberg', 'Pankow Süd': 'Pankow Süd', 'Fennpfuhl': 'Fennpfuhl', 'Teltower Damm': 'Teltower Damm'
Nord': 'Marzahn Nord', 'Osloer Straße': 'Osloer Straße', 'Prenzlauer Berg Süd': 'Prenzlauer Berg Süd
Viertel': 'Allende-Viertel', 'Lankwitz': 'Lankwitz', 'West 5': 'West 5 - Tegel', 'Schmargendorf': 'S
Lichtenberg': 'Alt-Lichtenberg', 'Haselhorst': 'Haselhorst', 'Neu-Hohenschönhausen Nord': 'Neu-
Hohenschönhausen Nord', 'West 4': 'West 4 - Auguste-Viktoria-Allee', 'Britz': 'Britz', 'Frankfurter
Reinickendorf', 'Nord 2': 'Nord 2 - Waidmannslust/Wittenau/Lübars', 'Wilhelmstadt': 'Wilhelmstadt',
Marx-Allee-Süd': 'Karl-Marx-Allee Süd', 'Marzahn-Süd': 'Marzahn Süd', 'Reuterstraße': 'Reuterstraße'
Nord': 'Hellersdorf Nord', 'Köpenick-Nord': 'Köpenick Nord', 'West 3': 'West 3 - Borsigwalde/Freie S
Hohenschönhausen Süd': 'Neu-Hohenschönhausen Süd', 'Nord 1': 'Nord 1 - Frohnau/Hermsdorf', 'Altglien
Treptow', 'Hakenfelde': 'Hakenfelde', 'Rudow': 'Rudow', 'Brunnenstr. Nord': 'Brunnenstraße Nord', 'W
Nord': 'Schöneberg Nordost', 'Schöneberg-Süd': 'Schöneberg Südost', 'Kölln. Vorstadt/Spindlersf.': '
```

A few unmatched regions remain after the name-based matching. In principle, these cases can be handled more reliably by using the listing coordinates and assigning each point directly to its geographic polygon. This is the better approach because it does not depend on naming conventions.

```
import geopandas as gpd

# Build a GeoDataFrame from the listing coordinates.
listings_1_11 = listings_1_9.copy()

gdf_listings = gpd.GeoDataFrame(
    listings_1_11,
    geometry=gpd.points_from_xy(listings_1_11["longitude"], listings_1_11["latitude"]),
    crs="EPSG:4326"
)

# Load the GeoJSON file.
gdf_bezirke = gpd.read_file("../Daten/berlin_bezirke.geojson")
gdf_bezirke = gdf_bezirke.to_crs(gdf_listings.crs)

# Spatial join: assign each listing to its Bezirksregion.
```

```
result = gpd.sjoin(
    gdf_listings,
    gdf_bezirke[["BZR_NAME", "GROESSE_m2", "geometry"]],
    how="left",
    predicate="within"
)

# Copy the matched region information back to the main table.
listings_1_11[["BZR_NAME", "GROESSE_m2"]] = result[["BZR_NAME", "GROESSE_m2"]].values
listings_1_11 = listings_1_11.drop(columns=["neighbourhood_cleansed"])
listings_1_11.head()
```

	price	latitude	longitude	room_type	bedrooms	bathrooms_number	accommodates	amenities
0	105.0	52.53471	13.41810	Entire home/apt	1.0	1.0	2	["Cooking ba
1	135.0	52.53269	13.41805	Entire home/apt	4.0	2.5	7	["Cooking ba
2	75.0	52.54813	13.40366	Entire home/apt	0.0	1.0	1	["Ethernet co
3	32.0	52.49419	13.42166	Entire home/apt	0.0	1.0	2	["Cooking ba
4	202.0	52.53711	13.40888	Entire home/apt	2.0	1.0	4	["Cooking ba

```
# Task 1.12

listings_1_12 = listings_1_11.loc[listings_1_11["price"] < upper_bound].copy()

region_stats = (
    listings_1_12
    .groupby("BZR_NAME")
    .agg(
        listings=("BZR_NAME", "count"),
        median_price=("price", "median")
    )
    .reset_index()
)

import plotly.express as px

fig = px.choropleth(
    region_stats,
    geojson=berlin_geojson,
    locations="BZR_NAME",
    featureidkey="properties.BZR_NAME",
    color="median_price",
    hover_name="BZR_NAME",
    hover_data={"listings": True, "median_price": True},
    title="Median nightly price of Airbnb listings by Bezirksregion in Berlin"
)

fig.update_geos(fitbounds="locations", visible=True)
fig.update_layout(height=550, margin=dict(l=0, r=0, t=50, b=0))
fig.show()

# Task 1.13
listings_1_13 = listings_1_12.copy()
listings_1_13["region_median_price"] = listings_1_13.groupby("BZR_NAME")["price"].transform("median"
listings_1_13["above_local_median"] = listings_1_13["price"] > listings_1_13["region_median_price"]

scatter = px.scatter_geo(
```

```
    listings_1_13,
    lat="latitude",
    lon="longitude",
    color="above_local_median",
    hover_data=["price", "region_median_price", "BZR_NAME"]
)

for trace in scatter.data:
    fig.add_trace(trace)
    fig.update_traces(marker=dict(size=5, opacity=0.5), selector=dict(mode='markers'))

fig.update_layout(height=550, margin=dict(l=0, r=0, t=50, b=0), legend={"x": 0.94, "y": 1})
fig.show()
```

```
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```

```
Unable to display output for mime type(s): text/html
```

```
Unable to display output for mime type(s): text/html
```

```
# Task 1.14

region_stats.head()

region_stats = region_stats.merge(gdf_bezirke[["BZR_NAME", "GROESSE_m2"]], on="BZR_NAME", how="left"
region_stats["GROESSE_km2"] = (region_stats["GROESSE_m2"] / 1_000_000).astype(float)
region_stats["listings_per_km2"] = (region_stats["listings"] / region_stats["GROESSE_km2"]).astype(fl

fig = px.choropleth(
    region_stats,
    geojson=berlin_geojson,
    locations="BZR_NAME",
    featureidkey="properties.BZR_NAME",
    color="listings_per_km2",
    hover_name="BZR_NAME",
    hover_data={"listings": True, "median_price": True},
    title="Airbnb listing density by Bezirksregion in Berlin"
)
fig.update_geos(fitbounds="locations", visible=True)
fig.update_layout(height=550, margin=dict(l=0, r=0, t=50, b=0))
fig.show()
```

```
Unable to display output for mime type(s): text/html
```

Listing density is clearly highest in central Berlin. The density map shows a strong concentration of Airbnb activity near the inner city, while most outer areas have far fewer listings per square kilometer. The price map points in a similar direction, but the pattern is less uniform. Central areas are often expensive, yet some peripheral regions also show high median prices. These cases should be interpreted carefully, because regions with only a small number of listings are more affected by individual high-price observations.

Taken together, the maps suggest that centrality matters for both density and price, but more strongly for density. Proximity to major attractions, public transport, and established urban neighborhoods likely explains why the centre dominates the market. At the same time, local factors such as neighborhood image, housing type, or small sample sizes can create high-price pockets outside the centre.

The following markdown cell was kept in the original notebook as a draft note. The final interpretation is given in the previous cell.

The maps suggest a clear centre–periphery pattern. Listing density is much higher in central Berlin than in the outer parts of the city. This is plausible because central areas are closer to tourist attractions, nightlife, restaurants, and major transport links. Median prices also tend to be higher in the centre, but

the pattern is less clean than for density. Some non-central Bezirksregionen show relatively high median prices as well. These values should be read carefully, since they may be driven by a limited number of listings rather than a broad market pattern. Overall, centrality seems to matter strongly for where Airbnb listings are concentrated, while prices are shaped by a mix of location, local demand, and listing-specific features such as size and amenities.

```
#listings_1_13.to_csv("listings_1_13.csv")
```

Section 2

```
# Task 2.1
# Load the cleaned data from Section 1 and prepare the regression sample.
from pathlib import Path
import pandas as pd
import numpy as np

if 'listings_1_13' in globals():
    listings_2 = listings_1_13.copy()
else:
    candidate_paths = [
        Path("../Aufgabe 1/listings_1_13.csv"),
        Path("../Daten/listings_1_13.csv"),
        Path("listings_1_13.csv")
    ]

    for path in candidate_paths:
        if path.exists():
            listings_2 = pd.read_csv(path, index_col=0)
            break
    else:
        raise FileNotFoundError("Could not find listings_1_13.csv in the expected relative paths.")

listings_2["price"] = listings_2["price"].replace(0, pd.NA)
listings_2.dropna(subset=["price"], inplace=True)

print(listings_2.columns)

listings_2_1 = listings_2[[
    "price", "accommodates", "latitude", "longitude", "amenities_count",
    "bedrooms", "host_is_superhost", "Shared_bath_status",
    "instant_bookable", "bathrooms_number"
]].copy()
listings_2_1.dropna(inplace=True)
listings_2_1["price"] = listings_2_1["price"].apply(np.log)
listings_2_1.rename(columns={"price": "log_price"}, inplace=True)

# Report the sample size used in the regression.
print(len(listings_2_1))
```

```
Index(['price', 'latitude', 'longitude', 'room_type', 'bedrooms',
       'bathrooms_number', 'accommodates', 'amenities', 'host_is_superhost',
       'host_listings_count', 'host_response_rate', 'review_scores_rating',
       'review_scores_cleanliness', 'number_of_reviews', 'reviews_per_month',
       'minimum_nights', 'availability_365', 'instant_bookable',
       'Shared_bath_status', 'host_tenure_years', 'amenities_list',
       'amenities_count', 'BZR_NAME', 'GROESSE_m2', 'region_median_price',
       'above_local_median'],
      dtype='str')
9093
```

The predictor set is based on the patterns from Section 1. We include accommodation size, bedrooms, bathrooms, amenities, location, and booking-related host characteristics. These variables capture the main dimensions that are likely to influence Airbnb prices: listing size, quality, convenience, and market position.

```
# Replace latitude and longitude with distance to the dataset's geographic center.
# This gives one compact location measure instead of two coordinates.
```

```
lon_center = listings_2_1["longitude"].mean()
lat_center = listings_2_1["latitude"].mean()

listings_2_1["distance_center"] = np.sqrt(
    (listings_2_1["latitude"] - lat_center) ** 2 +
    (listings_2_1["longitude"] - lon_center) ** 2
)

listings_2_1 = listings_2_1.drop(columns=["latitude", "longitude"])
```

```
#Aufgabe_2_2
import statsmodels.api as sm
outcome = listings_2_1["log_price"]
regressor = listings_2_1.drop(columns=["log_price"])
regressor = sm.add_constant(regressor)

bool_cols = regressor.select_dtypes("bool").columns
regressor[bool_cols] = regressor[bool_cols].astype(int)
#print(regressor.dtypes)
olssim = sm.OLS(outcome, regressor)
olssim = olssim.fit()
olssim.summary()
```

Dep. Variable:	log_price	**R-squared:**	0.465
Model:	OLS	**Adj. R-squared:**	0.464
Method:	Least Squares	**F-statistic:**	985.4
Date:	Fri, 20 Mar 2026	**Prob (F-statistic):**	0.00
Time:	12:59:28	**Log-Likelihood:**	-5265.1
No. Observations:	9093	**AIC:**	1.055e+04
Df Residuals:	9084	**BIC:**	1.061e+04
Df Model:	8		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.0318	0.017	236.237	0.000	3.998	4.065
accommodates	0.0770	0.003	23.520	0.000	0.071	0.083
amenities_count	0.0066	0.000	19.610	0.000	0.006	0.007
bedrooms	0.1199	0.008	14.763	0.000	0.104	0.136
host_is_superhost	0.0451	0.010	4.445	0.000	0.025	0.065
Shared_bath_status	-0.4623	0.013	-35.619	0.000	-0.488	-0.437
instant_bookable	0.1926	0.010	19.835	0.000	0.174	0.212
bathrooms_number	0.0896	0.011	8.083	0.000	0.068	0.111
distance_center	-1.5488	0.093	-16.632	0.000	-1.731	-1.366

Omnibus:	343.010	**Durbin-Watson:**	1.659
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	1004.417
Skew:	-0.088	**Prob(JB):**	7.83e-219
Kurtosis:	4.619	**Cond. No.**	699.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Very expensive listings should not be removed only to make the model look cleaner. They should only be excluded if there is a substantive reason, for example if they are implausible or clearly outside the market segment we want to study. In our case, it is more sensible to treat them as part of the data and then check whether results are sensitive to their presence.

```
#cutoff = listings_2_1["log_price"].quantile(0.99)
#df_trim = listings_2_1.loc[listings_2_1["log_price"] <= cutoff].copy()
```

```
outcome = listings_2_1["log_price"]
regressor = sm.add_constant(listings_2_1.drop(columns=["log_price"]))
bool_cols2 = regressor.select_dtypes("bool").columns
regressor[bool_cols2] = regressor[bool_cols2].astype(int)
olssim = sm.OLS(outcome, regressor).fit()

#print(f"\nOLS excluding top {(1-p)*100:.1f}% of prices")
#print(f"Cutoff (log_price): {cutoff:.4f}")
#print(f"Observations: {len(df_trim)}")
olssim.summary()
```

Dep. Variable:	log_price	**R-squared:**	0.465
Model:	OLS	**Adj. R-squared:**	0.464
Method:	Least Squares	**F-statistic:**	985.4
Date:	Fri, 20 Mar 2026	**Prob (F-statistic):**	0.00
Time:	12:59:28	**Log-Likelihood:**	-5265.1
No. Observations:	9093	**AIC:**	1.055e+04
Df Residuals:	9084	**BIC:**	1.061e+04
Df Model:	8		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.0318	0.017	236.237	0.000	3.998	4.065
accommodates	0.0770	0.003	23.520	0.000	0.071	0.083
amenities_count	0.0066	0.000	19.610	0.000	0.006	0.007
bedrooms	0.1199	0.008	14.763	0.000	0.104	0.136
host_is_superhost	0.0451	0.010	4.445	0.000	0.025	0.065
Shared_bath_status	-0.4623	0.013	-35.619	0.000	-0.488	-0.437
instant_bookable	0.1926	0.010	19.835	0.000	0.174	0.212
bathrooms_number	0.0896	0.011	8.083	0.000	0.068	0.111
distance_center	-1.5488	0.093	-16.632	0.000	-1.731	-1.366

Omnibus:	343.010	**Durbin-Watson:**	1.659
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	1004.417
Skew:	-0.088	**Prob(JB):**	7.83e-219
Kurtosis:	4.619	**Cond. No.**	699.

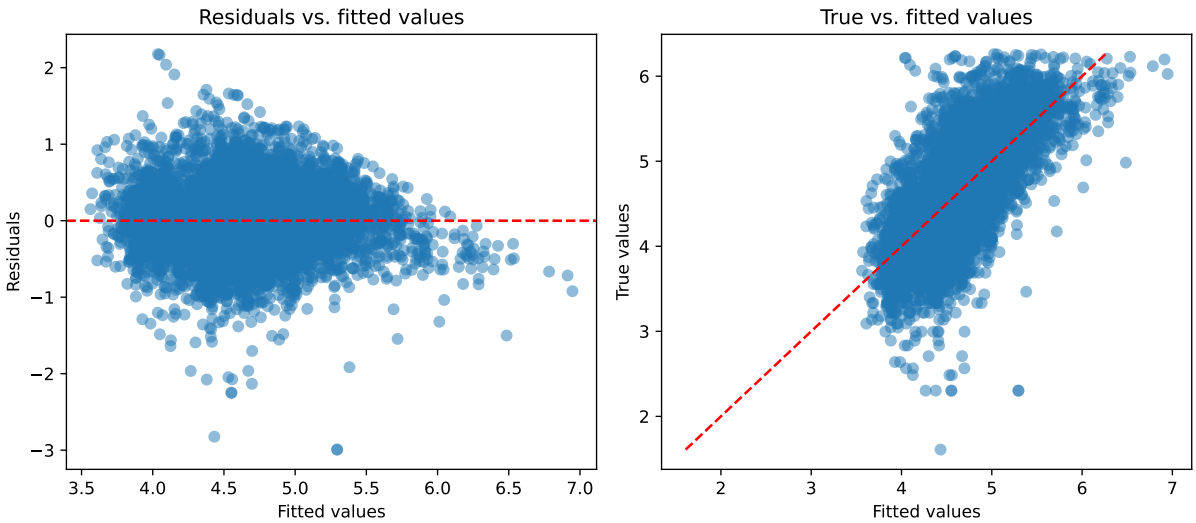
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The comparison suggests that extreme luxury listings affect the residual distribution and overall fit. Once the very top end is excluded, the model becomes somewhat more stable and the diagnostics improve. Still, this should be interpreted as a robustness check rather than the main specification, because those listings are part of the observed market.

A small p-value in a normality test would only speak against normally distributed residuals. It does **not** by itself prove a superhost premium. Any conclusion about a premium has to come from the estimated superhost coefficient and its statistical significance.

```
# Task 2.3

import matplotlib.pyplot as plt

fitted = olssim.fittedvalues
residuals = olssim.resid

fig, ax = plt.subplots(1, 2, figsize=(10, 4.5))

ax[0].scatter(fitted, residuals, alpha=0.5)
ax[0].axhline(0, color='red', linestyle='--')
ax[0].set_xlabel("Fitted values")
```

```
ax[0].set_ylabel("Residuals")
ax[0].set_title("Residuals vs. fitted values")

ax[1].scatter(fitted, outcome, alpha=0.5)
ax[1].plot([outcome.min(), outcome.max()], [outcome.min(), outcome.max()], 'r--')
ax[1].set_xlabel("Fitted values")
ax[1].set_ylabel("True values")
ax[1].set_title("True vs. fitted values")

plt.tight_layout()
plt.show()
print(len(outcome))
```



9093

The residuals are centered around zero and do not show a strong systematic pattern, which supports the linear specification as a reasonable first approximation. At the same time, there is still some spread and a few visible outliers, so the model should not be treated as perfect. The fitted-vs-true plot shows that the model captures broad price differences, but it does not explain all variation in listing prices.

```
#Aufgabe 2_4

import sklearn.linear_model
# Initiate linear regression model
#from sklearn.linear_model import LinearRegression
#model = LinearRegression()


ridgecv_reg = sklearn.linear_model.RidgeCV(cv=10, fit_intercept = False)
model_ridge = ridgecv_reg.fit(regressor, outcome)

print(model_ridge.coef_)
Y2_est = model_ridge.predict(regressor) # Prediction for Y2
MSE = np.mean((outcome-Y2_est)**2)
print("Mean-Squared-Error:", MSE)
model_ridge.score(regressor, outcome)
```

```
[ 4.03070231  0.07701957  0.00659014  0.11985903  0.04518235 -0.46216463
  0.19271042  0.08980038 -1.54034958]
Mean-Squared-Error: 0.1864068987549124
```

```
0.46460813265601997
```

```python
# Task 2.4: manual Ridge estimation with NumPy
def ridge_estimate(X: np.ndarray, y: np.ndarray, lam: float) -> np.ndarray:
    """Compute Ridge coefficients using the closed-form formula."""
    X = np.asarray(X)
    y = np.asarray(y)
    p = X.shape[1]
    I = np.eye(p)

    beta_ridge = np.linalg.inv(X.T @ X + lam * I) @ (X.T @ y)
    return beta_ridge

def l2_norm(beta: np.ndarray) -> float:
    """Compute the Euclidean norm of the coefficient vector."""
    return np.sqrt(np.sum(beta ** 2))

lambdas = [0.01, 1, 100]
X_data = regressor.values
y_data = outcome.values

for lam in lambdas:
    beta_hat = ridge_estimate(X_data, y_data, lam)
    norm = l2_norm(beta_hat)

    print(f"Results for lambda = {lam}:")
    print(f"||beta||_2 (l2 norm): {norm:.6f}")
    print(f"Coefficients (excerpt): {beta_hat[:3]}")
    print("-" * 30)
```

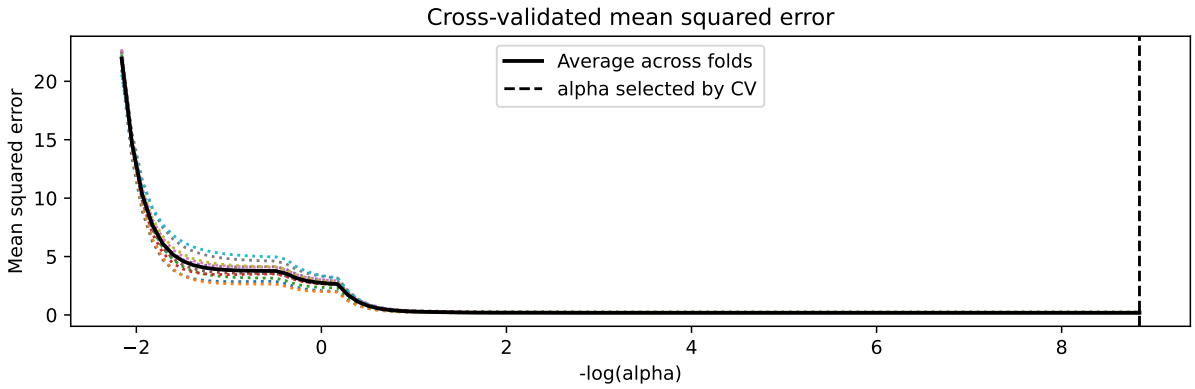
```
Results for lambda = 0.01:
||beta||_2 (l2 norm): 4.351098
Coefficients (excerpt): [4.03171157 0.07699538 0.0065857 ]
------------------------------
Results for lambda = 1:
||beta||_2 (l2 norm): 4.313049
Coefficients (excerpt): [4.02085447 0.07725541 0.00663449]
------------------------------
Results for lambda = 100:
||beta||_2 (l2 norm): 3.513462
Coefficients (excerpt): [3.47407698 0.0904675  0.01103062]
------------------------------
```

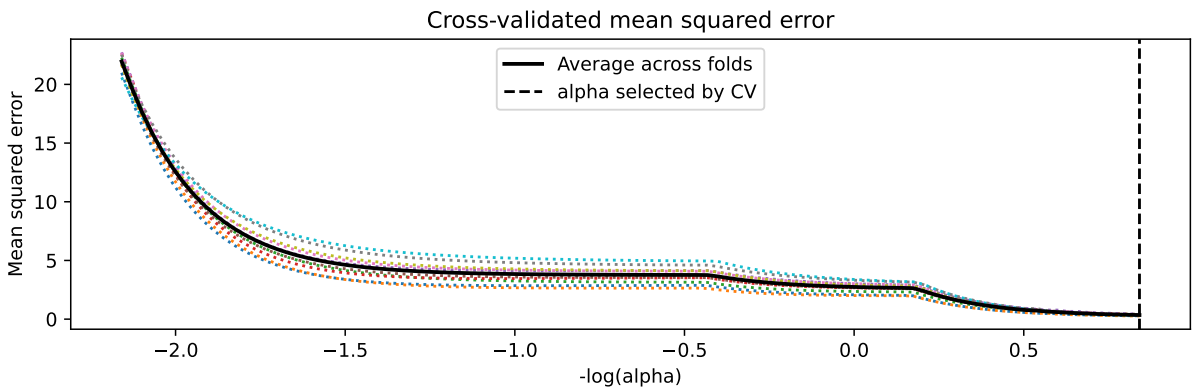
As expected, stronger Ridge regularization reduces the size of the coefficient vector. This is exactly the purpose of the penalty term: it shrinks coefficients toward zero and can stabilize the model when predictors are correlated. Compared with OLS, Ridge keeps all variables in the model but limits how strongly individual coefficients can move.

```python
# Task 2.5
# fit_intercept=False is used because the regressor matrix already contains a constant.
fig, ax = plt.subplots(2, 1, figsize=(9, 6))

def lasso_with_eps(epsilon, x_ax):
    lasso_reg = sklearn.linear_model.LassoCV(cv=10, fit_intercept=False, eps=epsilon)
    model = lasso_reg.fit(regressor, outcome)

    print("Lasso coefficients:")
    print(model.coef_)
    print("Eps:", model.eps)

    m_log_alphas = -np.log10(model.alphas_)
```

```
    ax[x_ax].plot(m_log_alphas, model.mse_path_, ':')
    ax[x_ax].plot(m_log_alphas, model.mse_path_.mean(axis=-1), 'k',
                  label='Average across folds', linewidth=2)
    ax[x_ax].axvline(-np.log10(model.alpha_), linestyle='--', color='k',
                     label='alpha selected by CV')
    ax[x_ax].legend()
    ax[x_ax].set_xlabel('-log(alpha)')
    ax[x_ax].set_ylabel('Mean squared error')
    ax[x_ax].set_title('Cross-validated mean squared error')
    ax[x_ax].axis('tight')

    zero_mask = model.coef_ == 0
    zero_features = regressor.columns[zero_mask]
    non_zero_features = regressor.columns[~zero_mask]

    print("Alpha selected via cross-validation:", model.alpha_)
    print("Zero-coefficient features:")
    print(list(zero_features))
    print("Non-zero-coefficient features:")
    print(list(non_zero_features))
    print("L2 norm of coefficients:", np.linalg.norm(model.coef_, 2))

lasso_with_eps(1e-11, 0)
lasso_with_eps(1e-3, 1)
plt.tight_layout()
plt.show()
```

```
Lasso coefficients:
[ 4.03182394  0.07699269  0.0065852   0.11988274  0.04511646 -0.462328
  0.19259124  0.08959394 -1.54878545]
Eps: 1e-11
Alpha selected via cross-validation: 1.4420211521492163e-09
Zero-coefficient features:
[]
Non-zero-coefficient features:
['const', 'accommodates', 'amenities_count', 'bedrooms', 'host_is_superhost', 'Shared_bath_status',
L2 norm of coefficients: 4.351503756299652
Lasso coefficients:
[ 3.05254322  0.18991629  0.02634185  0.          0.         -0.
  0.          0.03199987  0.        ]
Eps: 0.001
Alpha selected via cross-validation: 0.14420211521492163
Zero-coefficient features:
['bedrooms', 'host_is_superhost', 'Shared_bath_status', 'instant_bookable', 'distance_center']
Non-zero-coefficient features:
['const', 'accommodates', 'amenities_count', 'bathrooms_number']
L2 norm of coefficients: 3.058726239682447
```





With a very small epsilon, the selected alpha is also very small, so the Lasso solution stays close to OLS. In that case the regularization effect is weak and only limited coefficient shrinkage occurs. This suggests that the data do not strongly support a sparse model in which many predictors should be pushed exactly to zero.

```
#Aufgabe 2_6
from sklearn.model_selection import KFold
from sklearn.metrics import mean_squared_error
X = regressor.values
y = outcome.values
kf = KFold(n_splits=5, shuffle=True, random_state=12883823)
rmse_ols = []
rmse_ridge = []
rmse_lasso = []
for train, test in kf.split(X):
    X_train, X_test, y_train, y_test = X[train], X[test], y[train], y[test]

    # OLS
    ols_model_2_6 = sm.OLS(y_train, X_train).fit()
    y_pred_ols_2_6 = ols_model_2_6.predict(X_test)
    rmse_ols.append(np.sqrt(mean_squared_error(y_test, y_pred_ols_2_6)))

    # Ridge
    ridgecv_reg_2_6 = sklearn.linear_model.RidgeCV(cv=5, fit_intercept = False).fit(X_train, y_train
    Y2_est_2_6 = ridgecv_reg_2_6.predict(X_test)
    rmse_ridge.append(np.sqrt(mean_squared_error(y_test, Y2_est_2_6)))

    # Lasso
    lasso_reg_2_6 = sklearn.linear_model.LassoCV(cv=5,fit_intercept =False).fit(X_train, y_train)
    y_pred_lasso_2_6 = lasso_reg_2_6.predict(X_test)
    rmse_lasso.append(np.sqrt(mean_squared_error(y_test, y_pred_lasso_2_6)))

print("Average RMSE OLS:", np.mean(rmse_ols))
```

```
print("Average RMSE Ridge:", np.mean(rmse_ridge))
print("Average RMSE Lasso:", np.mean(rmse_lasso))
```

```
Average RMSE OLS: 0.43255376867743134
Average RMSE Ridge: 0.4325543567023152
Average RMSE Lasso: 0.5890117589685505
```

The cross-validated RMSE values are close to each other, but Lasso performs somewhat worse in this setting. This fits the structure of the problem: the predictor set is not especially sparse, so forcing coefficients toward zero can remove useful information. OLS and Ridge are therefore more natural benchmarks here, while Lasso is less attractive for prediction.

The regression results suggest that listing size and quality are central for pricing. Listings that accommodate more guests, offer more bedrooms and bathrooms, and provide more amenities tend to be more expensive. Location also matters: being closer to the city centre is generally associated with higher prices. In addition, host-related features such as superhost status and instant booking appear to be linked to a price premium, although these effects are smaller than the effects of size and location. For a new host, the main advice would be to benchmark the listing against similar nearby properties and then adjust for concrete characteristics such as capacity, equipment, and booking convenience. A host should not simply copy the average market price. Instead, pricing should reflect the local area, the size of the apartment, and the quality of the offering. Small improvements in presentation and amenities may justify a moderate premium, but large pricing mistakes are more likely to come from ignoring location and listing size.

```
# Task 2.8
model_2_8 = sm.QuantReg(outcome, regressor)
res_2_8 = model_2_8.fit(q=0.5)
res_2_8.summary().tables[1]
```

	coef	std err	t	P> |t|	[0.025	0.975]
const	4.0375	0.021	196.459	0.000	3.997	4.078
accommodates	0.0822	0.004	20.859	0.000	0.074	0.090
amenities_count	0.0066	0.000	16.349	0.000	0.006	0.007
bedrooms	0.1095	0.010	11.201	0.000	0.090	0.129
host_is_superhost	0.0466	0.012	3.811	0.000	0.023	0.071
Shared_bath_status	-0.4556	0.016	-29.152	0.000	-0.486	-0.425
instant_bookable	0.1924	0.012	16.457	0.000	0.169	0.215
bathrooms_number	0.0964	0.013	7.221	0.000	0.070	0.123
distance_center	-1.7936	0.112	-15.996	0.000	-2.013	-1.574

```
# Task 2.9
# H0: beta_host_is_superhost = 0
# H1: beta_host_is_superhost != 0
ols_2_9 = sm.OLS(outcome, regressor).fit(cov_type="HC3")
ols_2_9.summary().tables[1]

print(pd.concat([ols_2_9.params, ols_2_9.tvalues, ols_2_9.pvalues], axis=1,
                keys=["Coefficient", "t-Statistic", "p-Value"]).loc["host_is_superhost"])

print(ols_2_9.conf_int(0.05).rename(columns={0: "2.5%", 1: "97.5%"}).loc["host_is_superhost"])
print(f"Percentage premium for superhost listings: {(np.exp(ols_2_9.params['host_is_superhost']) - 1
```

```
Coefficient    0.045116
t-Statistic    4.580941
p-Value        0.000005
Name: host_is_superhost, dtype: float64
2.5%     0.025813
97.5%    0.064420
Name: host_is_superhost, dtype: float64
Percentage premium for superhost listings: 4.61%
```

Section 3

```python
# Task 3.1
import pandas as pd

# Load the training data.
X_train = pd.read_csv("../Daten//X_train.csv", index_col=0)
y_train = pd.read_csv("../Daten//y_train.csv", index_col=0)

n_obs, n_variables = X_train.shape
print(f"Observations: {n_obs}, Variables: {n_variables}")

default_fraction = y_train.mean().values[0]
print(f"Fraction of defaults: {default_fraction:.2%}")

summary_stats = X_train.describe().loc[['mean', 'std', 'min', 'max']]
print(summary_stats)
```

```
Observations: 5000, Variables: 20
Fraction of defaults: 35.02%
      annual_income        age  employment_years  debt_to_income  \
mean   50309.452058  43.626200          5.733020        0.166050
std    29349.199007  12.364216          4.166518        0.159139
min     5473.630000  22.000000          0.100000        0.007300
max   338861.400000  65.000000         40.000000        1.500000

      credit_score  credit_utilization  num_open_accounts  \
mean    678.794200            0.280992           4.948400
std      75.449405            0.156996           2.193422
min     434.000000            0.006200           0.000000
max     850.000000            0.869900          13.000000

      num_credit_inquiries   loan_amount  interest_rate  loan_term_months  \
mean              2.473600  11827.435264       8.402370         37.152000
std               1.568373   6533.911480       2.097997         14.089961
min               0.000000   1812.350000       2.000000         12.000000
max              10.000000  77433.770000      16.390000         60.000000

      home_ownership  months_since_delinquency  total_credit_lines  \
mean        0.415200                  8.968600            7.933600
std         0.492806                 16.673632            2.810973
min         0.000000                  0.000000            0.000000
max         1.000000                156.000000           21.000000

      revolving_balance  savings_balance  payment_history_score  \
mean       5.747878e+03      1804.155792              75.535580
std        2.421299e+04      2281.415292              13.619852
min        1.063000e+01        19.780000              16.800000
max        1.436268e+06     30942.310000             100.000000

      num_dependents  previous_defaults  monthly_installment
mean        0.988200           0.092400           436.871034
std         0.994715           0.289619           351.666441
min         0.000000           0.000000            53.670000
max         7.000000           1.000000          4595.210000
```

Distance-based methods such as KNN and SVM are sensitive to the scale of the input variables. Features like annual income, loan amount, and revolving balance have much larger numerical ranges than binary indicators or ratios. Without scaling, those large-scale variables would dominate the distance calculations.

For that reason, standardization is necessary before applying methods that rely on distances or margins in feature space.

```
# Task 3.2
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score

# Standardize the features.
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)

# Fit logistic regression.
log_reg = LogisticRegression()
log_reg.fit(X_train_scaled, y_train.values.ravel())

# Report training accuracy.
y_pred = log_reg.predict(X_train_scaled)
train_accuracy = accuracy_score(y_train, y_pred)
print(f"Training Accuracy: {train_accuracy:.4f}")

# Rank features by absolute coefficient size.
coef_df = pd.DataFrame({
    'Feature': X_train.columns,
    'Coefficient': log_reg.coef_[0],
    'Abs_Coefficient': abs(log_reg.coef_[0])
}).sort_values(by='Abs_Coefficient', ascending=False)

print("Top 3 features by absolute coefficient:")
print(coef_df.head(3))
```

```
Training Accuracy: 0.7298
Top 3 features by absolute coefficient:
               Feature  Coefficient  Abs_Coefficient
3       debt_to_income     0.853965         0.853965
5   credit_utilization     0.378353         0.378353
18   previous_defaults     0.368845         0.368845
```

After standardization, the coefficients can be compared more directly because all variables are measured on the same scale. A positive coefficient indicates that higher values of the feature are associated with a higher probability of default, while a negative coefficient points in the opposite direction. The variables with the largest absolute coefficients are therefore the strongest predictors in the logistic model.

```
# Task 3.3
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
import pandas as pd

# Split the data into training and validation sets.
X_train_k, X_val_k, y_train_k, y_val_k = train_test_split(
    X_train_scaled, y_train.values.ravel(), test_size=0.2, random_state=12883823
)

# Test different values of k and different weighting schemes.
k_values = [5, 20, 50]
weight_options = ['uniform', 'distance']

results = []

for k in k_values:
```

```
    for weight in weight_options:
        knn = KNeighborsClassifier(n_neighbors=k, weights=weight)
        knn.fit(X_train_k, y_train_k)

        accuracy = knn.score(X_val_k, y_val_k)
        error_rate = 1 - accuracy

        results.append({
            'k': k,
            'weights': weight,
            'error_rate': error_rate
        })

knn_results_df = pd.DataFrame(results)
print(knn_results_df)
```

```
    k   weights  error_rate
0   5   uniform       0.311
1   5  distance       0.311
2  20   uniform       0.312
3  20  distance       0.310
4  50   uniform       0.309
5  50  distance       0.308
```

KNN is strongly affected by the scale of the predictors because classification is based on distances between observations. After standardization, the method becomes more meaningful, but the results still depend on the choice of (k). Smaller values of (k) react more strongly to local variation, whereas larger values produce a smoother and more stable decision rule.

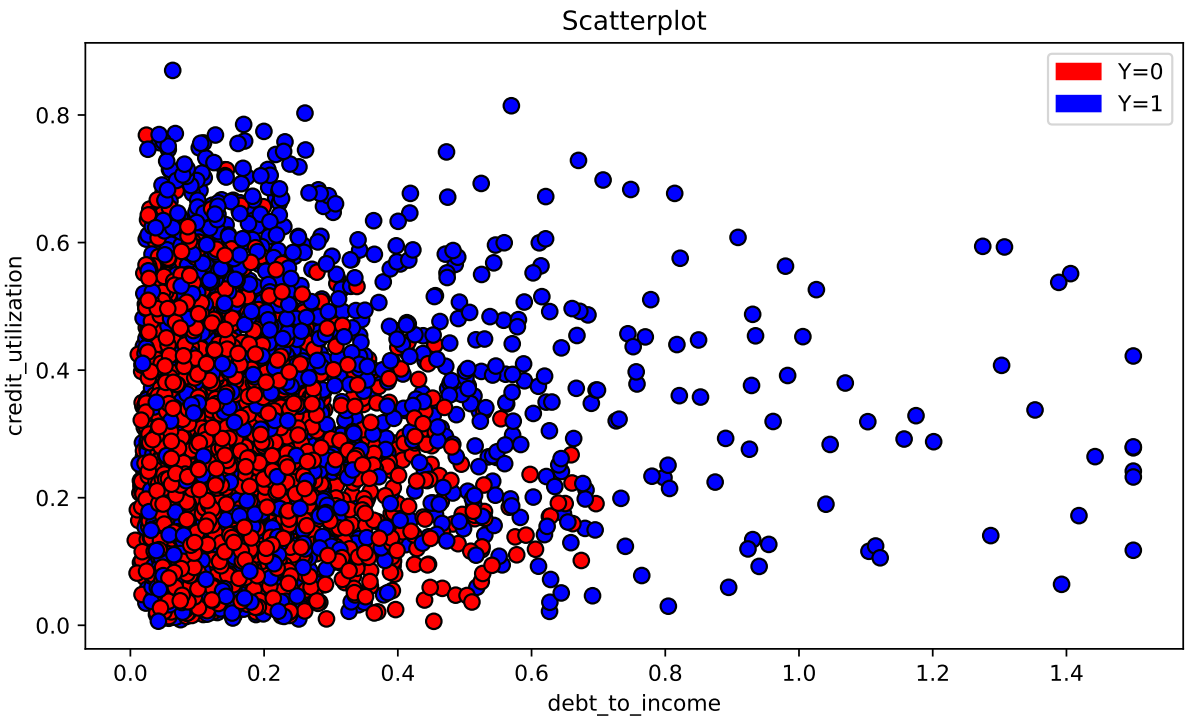
```
# Task 3.4
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Fit SVM with a linear kernel.
svm_linear = SVC(kernel='linear')
svm_linear.fit(X_train_k, y_train_k)
acc_linear = svm_linear.score(X_val_k, y_val_k)

# Fit SVM with an RBF kernel for different values of C.
c_values = [0.1, 1, 10]
rbf_results = {}

for c in c_values:
    svm_rbf = SVC(kernel='rbf', C=c)
    svm_rbf.fit(X_train_k, y_train_k)
    acc_rbf = svm_rbf.score(X_val_k, y_val_k)
    rbf_results[c] = acc_rbf

# Logistic regression benchmark.
log_reg_val = LogisticRegression()
log_reg_val.fit(X_train_k, y_train_k)
acc_log_reg = log_reg_val.score(X_val_k, y_val_k)

print(f"Logistic Regression Accuracy: {acc_log_reg:.4f}")
print(f"SVM Linear Kernel Accuracy: {acc_linear:.4f}")
for c, acc in rbf_results.items():
    print(f"SVM RBF Kernel (C={c}) Accuracy: {acc:.4f}")
```

```
Logistic Regression Accuracy: 0.7060
SVM Linear Kernel Accuracy: 0.6910
SVM RBF Kernel (C=0.1) Accuracy: 0.6970
SVM RBF Kernel (C=1) Accuracy: 0.7360
SVM RBF Kernel (C=10) Accuracy: 0.7060
```

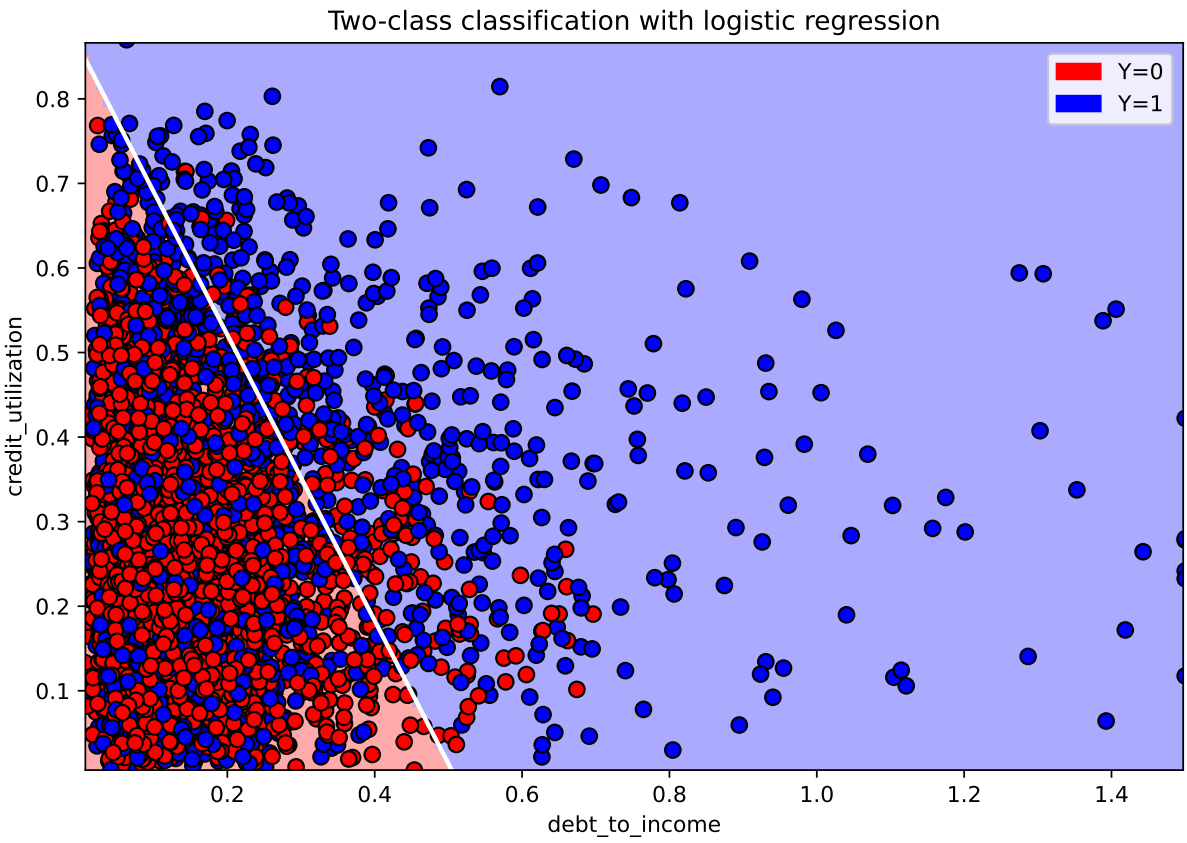
If the RBF kernel performs better than the linear kernel, this suggests that the relationship between the predictors and default risk is not purely linear. In that case, a curved decision boundary can separate the classes more effectively than a straight line. This would be plausible if default risk depends on interactions, nonlinear effects, or threshold behavior in the borrower characteristics.

```python
# Task 3.5
import numpy as np
X = X_train[["debt_to_income", "credit_utilization"]].values
y = y_train.values.ravel()

logisticRegr = LogisticRegression(l1_ratio=0, solver='liblinear')
model_3_5_1 = logisticRegr.fit(X, y)
y_score = model_3_5_1.predict(X)
print('Error rate:', np.mean(np.not_equal(model_3_5_1.predict(X), y)))

import matplotlib.pyplot as plt
from matplotlib.colors import ListedColormap
import matplotlib.patches as mpatches
plot_symbol_size = 50
cmap_bold = ListedColormap(['#FF0000', '#0000FF'])
plt.figure(figsize=(9, 5))
plt.scatter(X[:, 0], X[:, 1], s=plot_symbol_size, c=y, cmap=cmap_bold, edgecolor='black')
plt.xlabel("debt_to_income")
plt.ylabel("credit_utilization")
plt.title("Scatterplot")
patch0 = mpatches.Patch(color='#FF0000', label='Y=0')
patch1 = mpatches.Patch(color='#0000FF', label='Y=1')
plt.legend(handles=[patch0, patch1])
plt.show()
```

```
Error rate: 0.2874
```



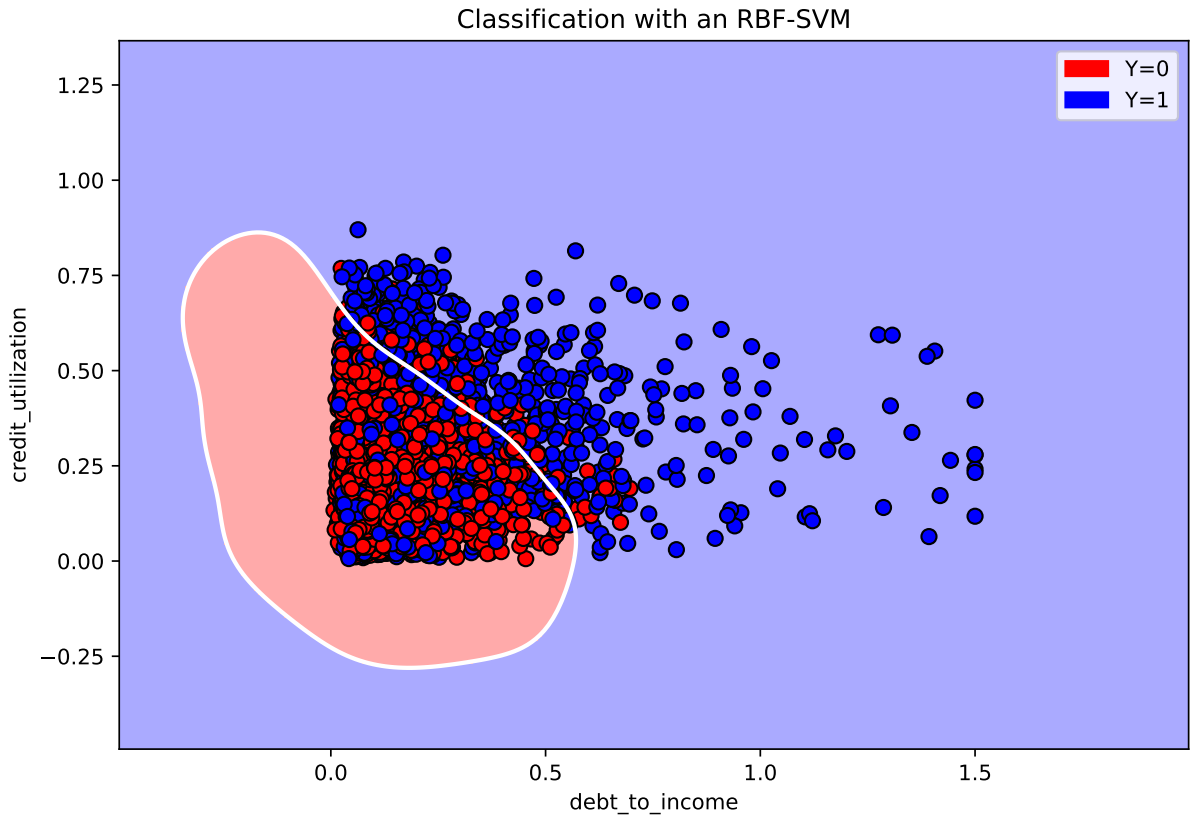

```
Y = y
pen = 'l2'
solv = 'liblinear'
# Create color maps.
cmap_light = ListedColormap(['#FFAAAA', '#AAAAFF'])
cmap_bold = ListedColormap(['#FF0000', '#0000FF'])

mesh_step_size = .01
plot_symbol_size = 50

x_min, x_max = X[:, 0].min(), X[:, 0].max()
y_min, y_max = X[:, 1].min(), X[:, 1].max()
xx, yy = np.meshgrid(np.arange(x_min, x_max, mesh_step_size),
                     np.arange(y_min, y_max, mesh_step_size))
Z = model_3_5_1.predict(np.c_[xx.ravel(), yy.ravel()])
D = model_3_5_1.decision_function(np.c_[xx.ravel(), yy.ravel()])
D = D.reshape(xx.shape)

Z = Z.reshape(xx.shape)
plt.figure(3, figsize=(9, 6))
plt.pcolormesh(xx, yy, Z, cmap=cmap_light)

plt.scatter(X[:, 0], X[:, 1], s=plot_symbol_size, c=Y, cmap=cmap_bold, edgecolor='black')
plt.xlim(xx.min(), xx.max())
plt.ylim(yy.min(), yy.max())

patch0 = mpatches.Patch(color='#FF0000', label='Y=0')
patch1 = mpatches.Patch(color='#0000FF', label='Y=1')
plt.legend(handles=[patch0, patch1])
plt.contour(xx, yy, D, levels=[0], colors='w', linestyles=['-'], linewidths=2)
plt.xlabel('debt_to_income')
plt.ylabel('credit_utilization')
plt.title("Two-class classification with logistic regression")
plt.show()
```




```
from sklearn import svm
supvecma = svm.SVC(kernel='rbf', C=3)
model_3_5_2 = supvecma.fit(X, Y)
print('Error rate:', np.mean(np.not_equal(model_3_5_2.predict(X), y)))
from matplotlib.colors import ListedColormap
import matplotlib.patches as mpatches
Y = y

cmap_light2 = ListedColormap(['#FFAAAA', '#AAAAFF'])
cmap_bold2 = ListedColormap(['#FF0000', '#0000FF'])

mesh_step_size = .01
plot_symbol_size = 50

x_min, x_max = X[:, 0].min() - .5, X[:, 0].max() + .5
y_min, y_max = X[:, 1].min() - .5, X[:, 1].max() + .5
xx, y_grid = np.meshgrid(np.arange(x_min, x_max, mesh_step_size),
                         np.arange(y_min, y_max, mesh_step_size))
Z = model_3_5_2.predict(np.c_[xx.ravel(), y_grid.ravel()])
D = model_3_5_2.decision_function(np.c_[xx.ravel(), y_grid.ravel()])
D = D.reshape(xx.shape)
Z = Z.reshape(xx.shape)

plt.figure(3, figsize=(9, 6))
plt.pcolormesh(xx, y_grid, Z, cmap=cmap_light2)
plt.scatter(X[:, 0], X[:, 1], s=plot_symbol_size, c=Y, cmap=cmap_bold2, edgecolor='black')
plt.xlim(xx.min(), xx.max())
plt.ylim(y_grid.min(), y_grid.max())

patch0 = mpatches.Patch(color='#FF0000', label='Y=0')
patch1 = mpatches.Patch(color='#0000FF', label='Y=1')
```

```
plt.legend(handles=[patch0, patch1])
plt.contour(xx, y_grid, D, levels=[0], colors='w', linestyles=['-'], linewidths=2)
plt.xlabel('debt_to_income')
plt.ylabel('credit_utilization')
plt.title("Classification with an RBF-SVM")
plt.show()
```

```
Error rate: 0.2748
```




```
# Task 3.6

import numpy as np
from sklearn.ensemble import RandomForestClassifier
randforest = RandomForestClassifier(max_depth=5, random_state=0)
model_3_6 = randforest.fit(X_train, y_train.values.ravel())
print("Error rate:", np.mean(model_3_6.predict(X_train) != y_train.values.ravel()))
importance = pd.DataFrame([model_3_6.feature_importances_], index=["importance"], columns=X_train.col
importance.head()
```

```
Error rate: 0.2366
```

	importance
credit_utilization	0.240613
debt_to_income	0.237807
previous_defaults	0.109675
annual_income	0.088058
payment_history_score	0.067746

The leading variables from the random forest importance ranking are broadly in line with the earlier logistic-regression results, even if the exact order differs. This is reassuring because two different model classes point to a similar set of important predictors.

```python
# Task 3.7a
from sklearn.model_selection import KFold
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import log_loss

# Use scaled predictors for the model comparison.
X = X_train_scaled
y = y_train.values.ravel()

kf = KFold(n_splits=5, shuffle=True, random_state=12883823)
acc_log_reg_3_7 = []
acc_knn_3_7 = []
acc_svm_3_7 = []
acc_rf_3_7 = []

ll_log_reg_3_7 = []
ll_knn_3_7 = []
ll_svm_3_7 = []
ll_rf_3_7 = []

for train, test in kf.split(X):
    X_train_3_7, X_test_3_7, y_train_3_7, y_test_3_7 = X[train], X[test], y[train], y[test]

    model_3_7_1 = LogisticRegression().fit(X_train_3_7, y_train_3_7)
    acc_log_reg_3_7.append(accuracy_score(y_test_3_7, model_3_7_1.predict(X_test_3_7)))
    ll_log_reg_3_7.append(log_loss(y_test_3_7, model_3_7_1.predict_proba(X_test_3_7)))

    model_3_7_2 = KNeighborsClassifier(n_neighbors=20, weights='distance').fit(X_train_3_7, y_train_3
    acc_knn_3_7.append(accuracy_score(y_test_3_7, model_3_7_2.predict(X_test_3_7)))
    ll_knn_3_7.append(log_loss(y_test_3_7, model_3_7_2.predict_proba(X_test_3_7)))

    model_3_7_3 = SVC(kernel='rbf', C=10, probability=True).fit(X_train_3_7, y_train_3_7)
    acc_svm_3_7.append(accuracy_score(y_test_3_7, model_3_7_3.predict(X_test_3_7)))
    ll_svm_3_7.append(log_loss(y_test_3_7, model_3_7_3.predict_proba(X_test_3_7)))

    model_3_7_4 = RandomForestClassifier(max_depth=5, random_state=0).fit(X_train_3_7, y_train_3_7)
    y_pred_3_7_4 = model_3_7_4.predict(X_test_3_7)
    acc_rf_3_7.append(accuracy_score(y_test_3_7, y_pred_3_7_4))
    ll_rf_3_7.append(log_loss(y_test_3_7, model_3_7_4.predict_proba(X_test_3_7)))

print(f"Average Accuracy Logistic Regression: {np.mean(acc_log_reg_3_7):.2%}")
print(f"Average Accuracy KNN: {np.mean(acc_knn_3_7):.2%}")
print(f"Average Accuracy SVM: {np.mean(acc_svm_3_7):.2%}")
print(f"Average Accuracy Random Forest: {np.mean(acc_rf_3_7):.2%}")
print(f"Average Log Loss Logistic Regression: {np.mean(ll_log_reg_3_7):.2f}")
print(f"Average Log Loss KNN: {np.mean(ll_knn_3_7):.2f}")
print(f"Average Log Loss SVM: {np.mean(ll_svm_3_7):.2f}")
print(f"Average Log Loss Random Forest: {np.mean(ll_rf_3_7):.2f}")
```

```
Average Accuracy Logistic Regression: 72.86%
Average Accuracy KNN: 71.74%
```

```
Average Accuracy SVM: 71.24%
Average Accuracy Random Forest: 73.70%
Average Log Loss Logistic Regression: 0.57
Average Log Loss KNN: 0.63
Average Log Loss SVM: 0.57
Average Log Loss Random Forest: 0.56
```

The cross-validated comparison points to Random Forest as the strongest model in this notebook. It achieves the best overall performance not only in terms of accuracy but also in terms of log loss. That matters because accuracy alone ignores how confident the predicted probabilities are. Two models can have the same number of correct classifications, while one produces much more sensible probability estimates. Looking at both measures therefore gives a fairer comparison. In our results, Random Forest performs best on both dimensions, so it is the most convincing choice for the final prediction task.

```python
# Task 3.8
X_test_3_8 = pd.read_csv("../Daten//X_test.csv", index_col=0)
print(X_test_3_8)
# Random Forest performed best in Task 3.7, so we use it for the final predictions.
y_test_pred = model_3_6.predict_proba(X_test_3_8)

model_3_6.classes_
predictions = y_test_pred[:, 1]
print(predictions)
pd.Series(predictions).to_csv("../Daten//predictions.csv", header=["probability"], index_label="id")
```

```
      annual_income   age  employment_years  debt_to_income  credit_score  \
5000       61147.74  31.0               5.6          0.1838         643.0
5001       52349.60  23.0               3.2          0.1138         708.0
5002       33045.87  53.0               5.2          0.1785         634.0
5003      108675.59  45.0               2.8          0.0517         642.0
5004       32740.92  40.0               1.8          0.1445         727.0
...             ...   ...               ...             ...           ...
7495       25029.34  36.0               2.0          0.1550         657.0
7496       50085.90  60.0               5.2          0.0828         745.0
7497       56853.36  57.0              10.8          0.0940         760.0
7498       41850.21  65.0               6.1          0.0846         702.0
7499       75377.50  31.0               4.3          0.1191         682.0

      credit_utilization  num_open_accounts  num_credit_inquiries  \
5000              0.2592                  5                     4
5001              0.2340                  5                     2
5002              0.5987                  7                     3
5003              0.4714                  3                     1
5004              0.1429                  9                     3
...                  ...                ...                   ...
7495              0.1679                  6                     1
7496              0.1610                  4                     3
7497              0.2588                  8                     1
7498              0.2760                  6                     3
7499              0.4547                  5                     4

      loan_amount  interest_rate  loan_term_months  home_ownership  \
5000     11384.48           6.58                60               1
5001     15651.77           7.89                36               0
5002      4808.37           7.57                48               0
5003     10038.49           9.85                24               1
5004     12438.32           6.75                36               0
...           ...            ...               ...             ...
7495      6927.42           8.74                60               1
7496     15795.46           6.56                60               1
```

```
7497     13668.83           7.03                36               0
7498     12976.85           9.71                60               1
7499     13719.46           3.96                24               0

      months_since_delinquency  total_credit_lines  revolving_balance  \
5000                       0.0                   7           35675.38
5001                       1.0                   5             344.44
5002                       0.0                  11           18755.41
5003                       6.0                   8             275.22
5004                       2.0                   9             585.23
...                        ...                 ...                ...
7495                       4.0                   9            9015.97
7496                       0.0                   6            1797.19
7497                      10.0                  12            1158.30
7498                       0.0                  10            1063.22
7499                       0.0                   8            7615.61

      savings_balance  payment_history_score  num_dependents  \
5000           547.75                   87.5               1
5001          6559.58                   63.7               2
5002           304.39                   78.6               2
5003          2741.28                   66.0               0
5004          2850.75                   55.2               1
...               ...                    ...             ...
7495          1173.62                   61.2               3
7496           566.63                   95.2               0
7497           764.20                   52.7               2
7498          1236.85                   42.7               0
7499          1559.44                   53.4               0

      previous_defaults  monthly_installment
5000                  0               223.18
5001                  0               489.68
5002                  0               116.42
5003                  0               462.53
5004                  1               382.64
...                 ...                  ...
7495                  0               142.93
7496                  0               309.50
7497                  0               422.24
7498                  1               273.87
7499                  1               595.52

[2500 rows x 20 columns]
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Declaration of Authenticity

"We/I hereby certify that We/I have independently prepared the foregoing paper without external assistance and have not used any aids other than those indicated in the attached list. All passages taken verbatim or in substance from publications have been marked as such. Furthermore, We/I certify that We/I have not previously submitted this paper in any other examination procedure."

Hamburg, 20.03.2026, Niclas Schwarzbach (1) _____

Place, Date, Signature (2) _____

Place, Date, Signature (3) _____

Place, Date, Signature (4) _____